CSE 535: Project 4 Report

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Please utilize this report to showcase the results **after** executing your code for each of the ten specified test cases. There are 2 test case files and in total 12 test cases. Test cases are numbered 1-1, 1-2, …, 1-6, 2-1, 2-2, …, 2-6. Remember to FLUSH the entire system state before processing set 2-1 of transactions (this means that all data maintained on both the server and clients must be cleared completely). Ensure that the results presented here align with those shown in your recording. If your system encounters a crash or fails to produce results for any of the items, please make a note of it clearly.

In showing the results of **PrintBalance()** function, you only need to fill out the white cells (i.e., the balance of a client on the respective servers).
In showing the results of **PrintDataStore()** function, you can only show the transaction blocks that have been added to the datastore in that particular test case (instead of printing the entire datastore).

Please finish the report by **Monday, December 16, 11:59 pm** and submit a **pdf file** using the provided link on Brightspace.

Test case 1-1

a) PrintBalance()

	100	501	1001	1650	2150	2800
S1	2	18				
S2	2	18				
S3	2	18				
S4	2	18				
S5			8	12		
S6			10	10		
S7			8	12		
S8			8	12		
S9					17	3
S10					17	3
S11					10	10
S12					17	3

b) PrintDataStore()

	DataStore
S1	["1 (100,501,8)"]
S2	["1 (100,501,8)"]
S3	["1 (100,501,8)"]
S4	["1 (100,501,8)"]
S5	["1 (1001,1650,2)"]
S6	[ ]
S7	["1 (1001,1650,2)"]
S8	["1 (1001,1650,2)"]
S9	["1 (2800,2150,7)"]
S10	["1 (2800,2150,7)"]
S11	[ ]
S12	["1 (2800,2150,7)"]

Test case 1-2

a) PrintBalance()

	1	101	299	301	1111	1201	1299
S1	14	1	6	19			
S2	14	1	6	19			
S3	14	1	6	19			
S4	14	1	6	19			
S5					15	2	13
S6					10	10	10
S7					15	2	13
S8					15	2	13
S9							
S10							
S11							
S12							

b) PrintDataStore()

	DataStore
S1	["2 (101,301,9)","3 (299,1,4)"]
S2	["2 (101,301,9)","3 (299,1,4)"]
S3	["2 (101,301,9)","3 (299,1,4)"]
S4	["2 (101,301,9)","3 (299,1,4)"]
S5	["2 (1201,1111,5)","3 (1201,1299,3)"]
S6	[ ]
S7	["2 (1201,1111,5)","3 (1201,1299,3)"]
S8	["2 (1201,1111,5)","3 (1201,1299,3)"]
S9	[ ]
S10	[ ]
S11	[ ]
S12	[ ]

Test case 1-3

a) PrintBalance()

	796	997	1890	1895	2990	2994	2995
S1	10	13					
S2	10	13					
S3	10	13					
S4	10	13					
S5			15	5			
S6			10	10			
S7			15	5			
S8			15	5			
S9					15	10	5
S10					15	10	5
S11					10	10	10
S12					15	10	5

b) PrintDataStore()

	DataStore
S1	["4 (791,997,3)"]
S2	["4 (791,997,3)"]
S3	["4 (791,997,3)"]
S4	["4 (791,997,3)"]
S5	["4 (1895,1890,5)"]
S6	[ ]
S7	["4 (1895,1890,5)"]
S8	["4 (1895,1890,5)"]
S9	["2 (2995,2990,5)"]
S10	["2 (2995,2990,5)"]
S11	[ ]
S12	["2 (2995,2990,5)"]

Test case 1-4

a) PrintBalance()

	196	197	1295	1990	2770	2799
S1	7	13				
S2	7	13				
S3	7	13				
S4	10	10				
S5			10	10		
S6			10	10		
S7			10	10		
S8			10	10		
S9					9	11
S10					9	11
S11					10	10
S12					9	11

b) PrintDataStore()

	DataStore
S1	["5 (196,197,3)"]
S2	["5 (196,197,3)"]
S3	["5 (196,197,3)"]
S4	[ ]
S5	[ ]
S6	[ ]
S7	[ ]
S8	[ ]
S9	["3 (2770,2799,1)"]
S10	["3 (2770,2799,1)"]
S11	[ ]
S12	["3 (2770,2799,1)"]

Test case 1-5

a) PrintBalance()

	333	691	707	973	1490	1495	1690	1695	2970	2975
S1	10	10	10	10						
S2	10	10	10	10						
S3	10	10	10	10						
S4	10	10	10	10						
S5					13	7	4	16		
S6					10	10	10	10		
S7					13	7	4	16		
S8					13	7	4	16		
S9									10	10
S10									10	10
S11									10	10
S12									10	10

b) PrintDataStore()

	DataStore
S1	[ ]
S2	[ ]
S3	[ ]
S4	[ ]
S5	["5 (1495,1490,3)","6 (1690,1695,6)"]
S6	[ ]
S7	["5 (1495,1490,3)","6 (1690,1695,6)"]
S8	["5 (1495,1490,3)","6 (1690,1695,6)"]
S9	[ ]
S10	[ ]
S11	[ ]
S12	[ ]

Test case 1-6

a) PrintBalance()

	333	777	888	973	1189	1415	2222	2333	2970	2975
S1	6	15	5	8						
S2	6	15	5	8						
S3	6	15	5	8						
S4	10	10	10	10						
S5					15	5				
S6					10	10				
S7					15	5				
S8					15	5				
S9							4	16	19	1
S10							4	16	19	1
S11							10	10	10	10
S12							4	16	19	1

b) PrintDataStore()

	DataStore
S1	["6 (973,707,2)","7 (333,691,4)","8 (888,777,5)"]
S2	["6 (973,707,2)","7 (333,691,4)","8 (888,777,5)"]
S3	["6 (973,707,2)","7 (333,691,4)","8 (888,777,5)"]
S4	[ ]
S5	["7 (1415,1189,5)"]
S6	[ ]
S7	["7 (1415,1189,5)"]
S8	["7 (1415,1189,5)"]
S9	["4 (2975,2970,9)","5 (2222,2333,6)"]
S10	["4 (2975,2970,9)","5 (2222,2333,6)"]
S11	[ ]
S12	["4 (2975,2970,9)","5 (2222,2333,6)"]

Test case 2-1

a) PrintBalance()

	299	1001	1111	1999	2150	2999
S1	7					
S2	7					
S3	7					
S4	7					
S5		4	19	13		
S6		4	19	13		
S7		10	10	10		
S8		4	19	13		
S9					1	16
S10					10	10
S11					1	16
S12					1	16

b) PrintDataStore()

	DataStore
S1	["1 P (299,1999,3)","2 C (299,1999,3)"]
S2	["1 P (299,1999,3)","2 C (299,1999,3)"]
S3	["1 P (299,1999,3)","2 C (299,1999,3)"]
S4	["1 P (299,1999,3)","2 C (299,1999,3)"]
S5	["1 P (299,1999,3)","2 C (299,1999,3)","3 P (1001,2999,6)","4 C (1001,2999,6)","5 P (2150,1111,9)","6 C (2150,1111,9)"]
S6	["1 P (299,1999,3)","2 C (299,1999,3)","3 P (1001,2999,6)","4 C (1001,2999,6)","5 P (2150,1111,9)","6 C (2150,1111,9)"]
S7	[ ]
S8	["1 P (299,1999,3)","2 C (299,1999,3)","3 P (1001,2999,6)","4 C (1001,2999,6)","5 P (2150,1111,9)","6 C (2150,1111,9)"]
S9	["1 P (1001,2999,6)","2 C (1001,2999,6)","3 P (2150,1111,9)","4 C (2150,1111,9)"]
S10	[ ]
S11	["1 P (1001,2999,6)","2 C (1001,2999,6)","3 P (2150,1111,9)","4 C (2150,1111,9)"]
S12	["1 P (1001,2999,6)","2 C (1001,2999,6)","3 P (2150,1111,9)","4 C (2150,1111,9)"]

Test case 2-2

a) PrintBalance()

	296	1997	2297	2593
S1	1			
S2	1			
S3	1			
S4	1			
S5		19		
S6		19		
S7		10		
S8		19		
S9			13	7
S10			10	10
S11			13	7
S12			13	7

b) PrintDataStore()

	DataStore
S1	["3 P (296,1997,1)","4 C (296,1997,1)","5 P (296,1997,8)","6 C (296,1997,8)"]
S2	["3 P (296,1997,1)","4 C (296,1997,1)","5 P (296,1997,8)","6 C (296,1997,8)"]
S3	["3 P (296,1997,1)","4 C (296,1997,1)","5 P (296,1997,8)","6 C (296,1997,8)"]
S4	["3 P (296,1997,1)","4 C (296,1997,1)","5 P (296,1997,8)","6 C (296,1997,8)"]
S5	["7 P (296,1997,1)","8 C (296,1997,1)","9 P (296,1997,8)","10 C (296,1997,8)"]
S6	["7 P (296,1997,1)","8 C (296,1997,1)","9 P (296,1997,8)","10 C (296,1997,8)"]
S7	[ ]
S8	["7 P (296,1997,1)","8 C (296,1997,1)","9 P (296,1997,8)","10 C (296,1997,8)"]
S9	["5 (2593,2297,3)"]
S10	[ ]
S11	["5 (2593,2297,3)"]
S12	["5 (2593,2297,3)"]

Test case 2-3

a) PrintBalance()

	793	1993	1998	2998
S1	3			
S2	3			
S3	3			
S4	3			
S5		17	10	
S6		17	10	
S7		10	10	
S8		17	10	
S9				10
S10				10
S11				10
S12				10

b) PrintDataStore()

	DataStore
S1	["7 P (793,1993,7)","8 C (793,1993,7)"]
S2	["7 P (793,1993,7)","8 C (793,1993,7)"]
S3	["7 P (793,1993,7)","8 C (793,1993,7)"]
S4	[“7 P (793,1993,7)","8 C (793,1993,7)"]
S5	[“11 P (793,1993,7)","12 C (793,1993,7)"]
S6	["11 P (793,1993,7)","12 C (793,1993,7)"]
S7	[ ]
S8	["11 P (793,1993,7)","12 C (793,1993,7)"]
S9	[ ]
S10	[ ]
S11	[ ]
S12	[ ]

Test case 2-4

a) PrintBalance()

	5	98	1333	1877	2333	2855
S1	8	12				
S2	8	12				
S3	8	12				
S4	8	12				
S5			10	10		
S6			10	10		
S7			10	10		
S8			10	10		
S9					10	10
S10					10	10
S11					10	10
S12					10	10

b) PrintDataStore()

	DataStore
S1	["9 (5,98,2)"]
S2	["9 (5,98,2)"]
S3	["9 (5,98,2)"]
S4	["9 (5,98,2)"]
S5	[ ]
S6	[ ]
S7	[ ]
S8	[ ]
S9	[ ]
S10	[ ]
S11	[ ]
S12	[ ]

Test case 2-5

a) PrintBalance()

	5	45	98	1333	1355	1877	2333	2855
S1	6	10	14					
S2	6	10	14					
S3	6	10	14					
S4	6	10	14					
S5				10	10	10		
S6				10	10	10		
S7				10	10	10		
S8				10	10	10		
S9							10	10
S10							10	10
S11							10	10
S12							10	10

b) PrintDataStore()

	DataStore
S1	[" 10 P (45,1355,5)","11 A (45,1355,5)","12 (5,98,2)"]
S2	["10 P (45,1355,5)","11 A (45,1355,5)","12 (5,98,2)"]
S3	["10 P (45,1355,5)","11 A (45,1355,5)","12 (5,98,2)"]
S4	["10P (45,1355,5)","11 A (45,1355,5)","12 (5,98,2)"]
S5	[ ]
S6	[ ]
S7	[ ]
S8	[ ]
S9	[ ]
S10	[ ]
S11	[ ]
S12	[ ]

Test case 2-6

a) PrintBalance()

	298	1061	1333	1355	1789	1877	2333	2476	2850	2855
S1	7									
S2	7									
S3	7									
S4	7									
S5		4	7	10	10	5				
S6		4	7	10	10	5				
S7		10	10	10	10	10				
S8		4	7	10	10	5				
S9							13	16	1	15
S10							10	10	10	10
S11							13	16	1	15
S12							13	16	1	15

b) PrintDataStore()

	DataStore
S1	["13 P (298,1789,3)","14 C (298,1789,3)"]
S2	["13 P (298,1789,3)","14 C (298,1789,3)"]
S3	["13 P (298,1789,3)","14 C (298,1789,3)"]
S4	["13 P (298,1789,3)","14 C (298,1789,3)"]
S5	[“13 P (298,1789,3)","14 C (298,1789,3)","15 P (1877,2855,5)","16 C (1877,2855,5)","17 P (1333,2333,3)","18 C (1333,2333,3)","19 P (2850,1234,9)","20 C (2850,1234,9)","21 P (1061,2476,6)","22 C (1061,2476,6)"]
S6	["12 C (793,1993,7)","13 P (298,1789,3)","14 C (298,1789,3)","15 P (1877,2855,5)","16 C (1877,2855,5)","17 P (1333,2333,3)","18 C (1333,2333,3)","19 P (2850,1234,9)","20 C (2850,1234,9)","21 P (1061,2476,6)","22 C (1061,2476,6)"]
S7	[ ]
S8	["12 C (793,1993,7)","13 P (298,1789,3)","14 C (298,1789,3)","15 P (1877,2855,5)","16 C (1877,2855,5)","17 P (1333,2333,3)","18 C (1333,2333,3)","19 P (2850,1234,9)","20 C (2850,1234,9)","21 P (1061,2476,6)","22 C (1061,2476,6)"]
S9	["6 P (1877,2855,5)","7 C (1877,2855,5)","8 P (1333,2333,3)","9 C (1333,2333,3)","10 P

	(2850,1234,9)","11 C (2850,1234,9)","12 P (1061,2476,6)","13 C (1061,2476,6)"]
S10	[ ]
S11	["6 P (1877,2855,5)","7 C (1877,2855,5)","8 P (1333,2333,3)","9 C (1333,2333,3)","10 P (2850,1234,9)","11 C (2850,1234,9)","12 P (1061,2476,6)","13 C (1061,2476,6)"]
S12	["6 P (1877,2855,5)","7 C (1877,2855,5)","8 P (1333,2333,3)","9 C (1333,2333,3)","10 P (2850,1234,9)","11 C (2850,1234,9)","12 P (1061,2476,6)","13 C (1061,2476,6)"]